

Chapter 7: AI Ethics and Recent Trends

Ethics, Security, and Emerging Technologies

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Outline

- 1 Responsible AI: Bias, Fairness, Explainability
- 2 Ethical and Societal Considerations
- 3 Privacy and Security in AI
- 4 Emerging Trends: Federated Learning, Embedded AI

What is Responsible AI?

Definition

Responsible AI refers to developing and deploying AI systems that are:

- Fair and unbiased
- Transparent and explainable
- Accountable and auditable
- Privacy-preserving and secure
- Beneficial to society

Why It Matters

As AI systems increasingly make decisions affecting human lives (hiring, lending, healthcare, criminal justice), ensuring responsibility is critical.

AI Bias: Understanding the Problem

What is AI Bias?

- Systematic errors in AI predictions
- Unfair outcomes for certain groups
- Reflects societal prejudices in data

Sources of Bias:

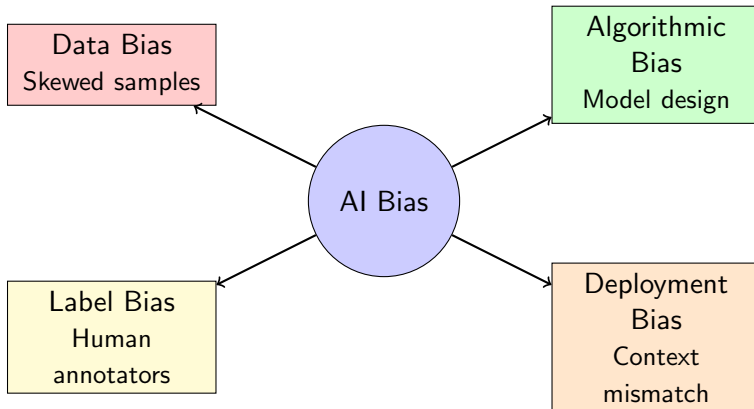
- 1 Historical bias in training data
- 2 Representation bias (undersampling)
- 3 Measurement bias
- 4 Algorithmic bias

Example

Real-world Cases:

- **COMPAS:** Racial bias in recidivism prediction
- **Amazon Recruiting:** Gender bias against women
- **Facial Recognition:** Higher error rates for darker skin tones
- **Word Embeddings:** Gender stereotypes (doctor=male, nurse=female)

Types of Bias in AI Systems



Fairness in Machine Learning

Fairness Metrics:

Individual Fairness

Similar individuals should receive similar predictions

$$d(x_i, x_j) < \epsilon \Rightarrow |f(x_i) - f(x_j)| < \delta$$

Demographic Parity

Predictions independent of protected attributes

$$P(\hat{Y} = 1 | A = a) = P(\hat{Y} = 1 | A = b)$$

Equal Opportunity

Equal true positive rates across groups

$$P(\hat{Y} = 1 | Y = 1, A = a) = P(\hat{Y} = 1 | Y = 1, A = b)$$

Equalized Odds

Equal TPR and FPR across groups

Challenge

These fairness definitions can be mathematically incompatible!

Bias Mitigation Techniques

Stage	Techniques
Pre-processing	• Re-sampling and re-weighting • Data augmentation for underrepresented groups • Learning fair representations
In-processing	• Adversarial debiasing • Regularization for fairness • Constraint-based optimization
Post-processing	• Threshold optimization • Calibration across groups • Reject option classification

Explainability: The Black Box Problem

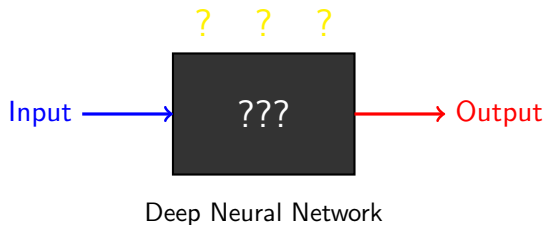
Why Explainability Matters:

- Trust and adoption
- Regulatory compliance (GDPR)
- Debugging and improvement
- Ethical accountability
- Safety in critical applications

Trade-off:

Accuracy vs Interpretability

- Linear models: Interpretable
- Deep neural networks: Accurate but opaque



Explainable AI (XAI) Techniques

1. Model-Agnostic Methods

- **LIME (Local Interpretable Model-agnostic Explanations):** Approximates complex models locally with interpretable ones
- **SHAP (SHapley Additive exPlanations):** Game-theoretic approach for feature importance
- **Partial Dependence Plots:** Shows relationship between features and predictions

2. Model-Specific Methods

- **Attention Mechanisms:** Highlights which input parts the model focuses on
- **Gradient-based Methods:** Saliency maps, Grad-CAM for CNNs
- **Decision Trees:** Inherently interpretable rules

SHAP Values: An Example

Shapley Values from Game Theory:

For a prediction $f(x)$, the SHAP value for feature i is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Example

House Price Prediction:

- Base prediction: \$250,000
- Feature contributions:
 - Square footage: +\$50,000
 - Location: +\$30,000
 - Age: -\$10,000
 - Number of rooms: +\$5,000
- Final prediction: \$325,000

AI Ethics: Core Principles

The Five Pillars

- 1 **Beneficence:** Do good
- 2 **Non-maleficence:** Do no harm
- 3 **Autonomy:** Respect human agency
- 4 **Justice:** Be fair
- 5 **Explicability:** Be transparent

Key Questions

- Who benefits from AI?
- Who is harmed?
- Who decides?
- Who is accountable?
- How do we ensure fairness?

IEEE Ethically Aligned Design

Guidelines for prioritizing human well-being in autonomous systems

Societal Impact of AI

Positive Impacts:

- Healthcare diagnostics
- Climate change modeling
- Accessibility tools
- Educational personalization
- Scientific discovery
- Disaster response

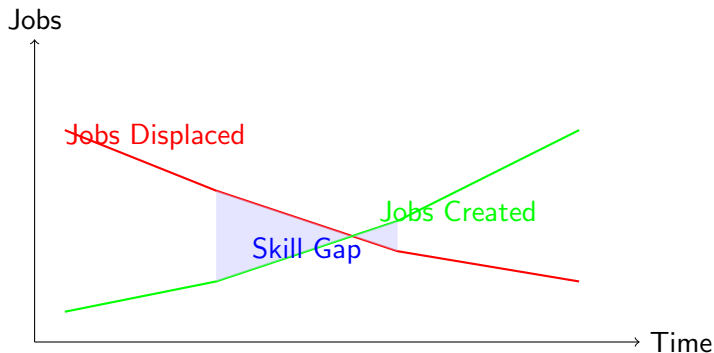
Concerns:

- Job displacement
- Algorithmic discrimination
- Surveillance and privacy
- Autonomous weapons
- Deepfakes and misinformation
- Digital divide and inequality

The Dual-Use Dilemma

Many AI technologies can be used for both beneficial and harmful purposes (e.g., facial recognition for unlocking phones vs. mass surveillance)

AI and Employment



Key Considerations:

- Routine tasks most vulnerable to automation
- Need for reskilling and upskilling programs
- Importance of human-AI collaboration
- Policy interventions: UBI, job guarantees, education reform

Algorithmic Decision-Making: Case Studies

Criminal Justice: COMPAS

- Risk assessment tool for recidivism
- ProPublica investigation found racial bias
- Black defendants labeled higher risk than white defendants
- Questions about opacity and accountability

Healthcare: Racial Bias in Algorithms

- Algorithm used healthcare costs as proxy for health needs
- Systematically underestimated needs of Black patients
- Historical disparities in healthcare spending perpetuated

Lesson

Seemingly neutral metrics can encode historical discrimination

Deepfakes and Misinformation

What are Deepfakes?

- AI-generated synthetic media
- Face swapping, voice cloning
- Based on GANs and autoencoders
- Increasingly realistic

Applications:

- Entertainment and art
- Education and training
- Accessibility (voice synthesis)

Risks:

- Political manipulation
- Non-consensual pornography
- Identity fraud
- Undermining trust in media
- Election interference

Countermeasures:

- Detection algorithms
- Digital watermarking
- Media literacy education
- Regulation and legislation

AI Governance and Regulation

European Union AI Act

Risk-based regulatory framework:

- **Unacceptable risk:** Banned (e.g., social scoring)
- **High risk:** Strict requirements (e.g., hiring, credit)
- **Limited risk:** Transparency obligations
- **Minimal risk:** No restrictions

Other Frameworks

- **GDPR:** Right to explanation for automated decisions
- **US Blueprint for AI Bill of Rights:** Five principles for AI protection
- **China's AI Regulations:** Algorithm registry, content control
- **UNESCO Recommendation on AI Ethics:** Global ethical framework

Privacy Challenges in AI

Key Privacy Risks:

- ➊ **Data Collection:** Massive datasets with personal information
- ➋ **Inference Attacks:** Learning sensitive info from model outputs
- ➌ **Model Inversion:** Reconstructing training data
- ➍ **Membership Inference:** Determining if data was in training set
- ➎ **Re-identification:** De-anonymizing datasets

Example

Netflix Prize Incident:

- Anonymized movie ratings dataset
- Researchers re-identified users by cross-referencing with IMDb
- Revealed political and sexual preferences
- Led to lawsuit and competition cancellation

Differential Privacy

Definition

A mathematical framework for privacy-preserving data analysis that provides provable privacy guarantees.

Formal Definition: A mechanism M is ϵ -differentially private if for all datasets D_1, D_2 differing in one record and all outputs S :

$$\frac{P[M(D_1) \in S]}{P[M(D_2) \in S]} \leq e^\epsilon$$

Key Ideas:

- Add calibrated noise to query results
- Privacy loss parameter ϵ (smaller = more private)
- Composition: Multiple queries accumulate privacy loss
- Used by: US Census, Apple, Google

Privacy-Preserving Machine Learning

1. Differential Privacy in ML

- Add noise during training (DP-SGD)
- Privacy budget management
- Trade-off: Privacy vs. Model accuracy

2. Homomorphic Encryption

- Computation on encrypted data
- $Enc(a) \oplus Enc(b) = Enc(a + b)$
- Still computationally expensive for deep learning

3. Secure Multi-Party Computation (MPC)

- Multiple parties compute without revealing inputs
- Secret sharing schemes
- Application: Collaborative learning without data sharing

Security Threats to AI Systems

Attack Type	Description
Adversarial Examples	Carefully crafted inputs that fool ML models (e.g., adding imperceptible perturbations to images)
Data Poisoning	Injecting malicious data into training sets to corrupt model behavior
Model Extraction	Stealing a proprietary model through query access
Backdoor Attacks	Embedding hidden triggers that activate malicious behavior
Evasion Attacks	Manipulating inputs to evade detection (e.g., spam filters, malware detectors)

Adversarial Examples

How They Work:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

- x : Original input
- ϵ : Perturbation magnitude
- J : Loss function
- Result: Misclassification with high confidence

Real-world Implications:

- Autonomous vehicles
- Face recognition systems
- Medical diagnosis
- Spam and malware detection

Example

FGSM Attack:

- Panda image: 57.7% confidence
- Add imperceptible noise
- Result: "Gibbon" with 99.3% confidence
- Humans cannot see difference

Defense Strategies:

- Adversarial training
- Input preprocessing
- Model ensembles
- Certified defenses

Data Poisoning Attacks

Attack Scenario

Attacker manipulates training data to degrade model performance or insert backdoors

Types:

- ① **Availability Attacks:** Reduce overall model accuracy
- ② **Targeted Attacks:** Misclassify specific inputs
- ③ **Backdoor Attacks:** Trigger specific behavior with special inputs

Example

Backdoor in Image Classifier:

- Inject training images with yellow square patch
- Label them as "stop sign"
- Deployed model: Normal images classified correctly
- Images with yellow patch: Always classified as "stop sign"
- Danger: Autonomous vehicle attacks

Securing AI Systems: Best Practices

Data Security:

- Data sanitization and validation
- Secure data storage and transmission
- Access control and auditing
- Data provenance tracking

Model Security:

- Adversarial training
- Input validation
- Model watermarking
- Regular security audits

Deployment Security:

- API rate limiting
- Query monitoring
- Model versioning
- Incident response plans

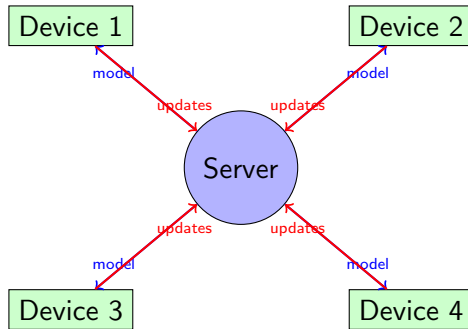
Organizational:

- Security-aware ML lifecycle
- Red team exercises
- Privacy impact assessments
- Compliance with regulations

Federated Learning: Overview

Definition

A distributed machine learning approach where model training occurs across decentralized devices holding local data, without exchanging the data itself.



Federated Learning: How It Works

FedAvg Algorithm (Federated Averaging):

- 1 Server sends global model w_t to selected clients
- 2 Each client k trains locally on their data:

$$w_t^k = w_t - \eta \nabla L_k(w_t)$$

- 3 Clients send updated weights back to server
- 4 Server aggregates updates:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_t^k$$

where n_k is the number of samples at client k

- 5 Repeat until convergence

Key Advantage: Data never leaves devices - privacy preserved!

Federated Learning: Applications & Challenges

Applications:

- **Mobile keyboards:** Google Gboard
- **Healthcare:** Hospital collaboration without data sharing
- **Finance:** Fraud detection across banks
- **IoT:** Edge device collaboration
- **Autonomous vehicles:** Cross-fleet learning

Challenges:

- **Non-IID data:** Heterogeneous data distributions
- **Communication costs:** Network bandwidth limitations
- **Device heterogeneity:** Varying computational capabilities
- **Privacy attacks:** Model inversion still possible
- **Convergence:** Slower than centralized training

Privacy Enhancement

Combine with differential privacy and secure aggregation for stronger guarantees

Variants of Federated Learning

Variant	Description
Horizontal FL	Clients have same features but different samples (e.g., hospitals with same patient data format)
Vertical FL	Clients have different features for same samples (e.g., bank and e-commerce company sharing user insights)
Federated Transfer Learning	Combines FL with transfer learning for different data distributions and feature spaces
Split Learning	Model split between client and server; only activations shared

Embedded AI: Bringing Intelligence to the Edge

What is Embedded AI?

Integration of AI/ML capabilities directly into embedded systems and edge devices, enabling on-device inference without cloud connectivity.

Key Technologies:

- TinyML: ML for microcontrollers
- Model compression
- Quantization
- Neural architecture search
- Specialized hardware (NPUs, TPUs)

Benefits:

- Low latency
- Privacy preservation
- Offline operation
- Reduced bandwidth
- Lower power consumption
- Cost efficiency

Model Optimization for Embedded Systems

1. Quantization

Reduce precision of weights and activations:

- FP32 $\rightarrow$ INT8 or even INT4
- 4x-8x reduction in model size
- Faster inference, lower power
- Minimal accuracy loss with proper techniques

2. Pruning

Remove unnecessary network connections:

- Structured pruning: Remove entire channels/layers
- Unstructured pruning: Remove individual weights
- Can achieve 90%+ sparsity with minimal accuracy loss

3. Knowledge Distillation

TinyML: Machine Learning on Microcontrollers

Characteristics:

- Models $\leq$ 1MB (often $\leq$ 100KB)
- Run on devices with KB of RAM
- Ultra-low power (milliwatts or less)
- Real-time inference

Popular Frameworks:

- TensorFlow Lite Micro
- Edge Impulse
- ARM CMSIS-NN
- STM32Cube.AI
- Apache TVM

Applications:

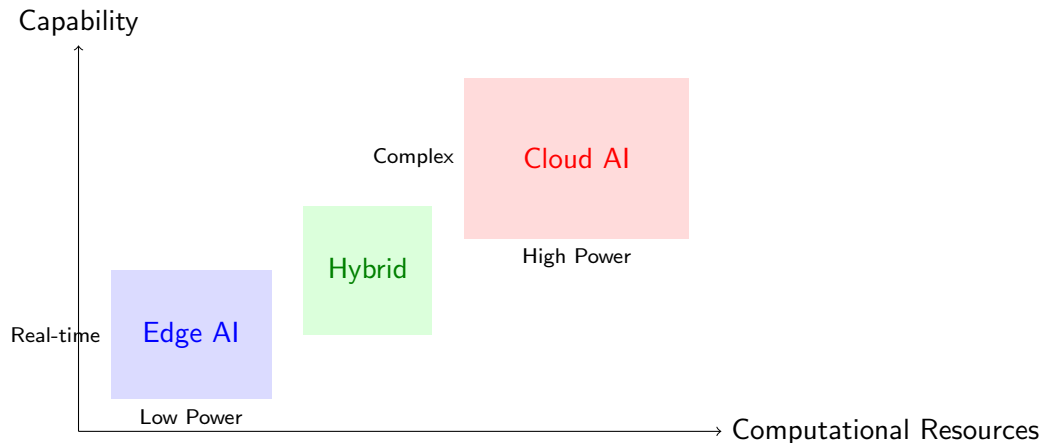
- Voice recognition (wake words)
- Gesture detection
- Anomaly detection in IoT
- Predictive maintenance
- Wildlife monitoring
- Health monitoring wearables

Hardware	Power	Use Cases
MCUs	mW	TinyML, sensors, wearables
Google Coral	0.5-2W	Embedded vision, robots
NVIDIA Jetson	5-30W	Autonomous systems, drones
Intel Movidius	1-2W	Cameras, AR/VR
Apple Neural Engine	Low	Smartphones, tablets
Qualcomm Hexagon	Low	Mobile devices, IoT

Design Considerations

Memory, power, latency, cost, thermal constraints all matter in embedded AI deployment

Edge AI vs Cloud AI



Hybrid Approach: Combine edge and cloud for optimal performance, privacy, and cost

Emerging Trends: Summary

1. Federated Learning

- Privacy-preserving collaborative learning
- Growing adoption in mobile and healthcare
- Active research: Byzantine robustness, personalization

2. Embedded AI / Edge AI

- Democratizing AI for resource-constrained devices
- TinyML enabling new IoT applications
- Trend: More capable models on smaller devices

3. Other Emerging Trends

- Neuromorphic computing (brain-inspired chips)
- Quantum machine learning
- AutoML and Neural Architecture Search

Key Takeaways

- ➊ **Responsible AI is Critical:** Bias, fairness, and explainability must be central to AI development
- ➋ **Ethics Matter:** AI's societal impact requires careful consideration of values, rights, and accountability
- ➌ **Privacy and Security are Paramount:** Protecting data and models against emerging threats is essential
- ➍ **Innovation Continues:** Federated learning and embedded AI are reshaping how we deploy AI systems
- ➎ **Engineers Have Responsibility:** As future AI practitioners, you shape the ethical landscape

Remember

"With great power comes great responsibility" - applies perfectly to AI!

Discussion Questions

- ➊ Should there be a global ban on certain AI applications (e.g., autonomous weapons, social scoring)? Why or why not?
- ➋ How can we balance the trade-off between model accuracy and fairness? Who should decide what "fair" means?
- ➌ Is complete explainability always necessary? When might we accept "black box" systems?
- ➍ How will federated learning change data privacy in healthcare and finance?
- ➎ What ethical responsibilities do you have as an AI engineer?

Resources for Further Learning

Books:

- "Weapons of Math Destruction" - Cathy O'Neil
- "The Alignment Problem" - Brian Christian
- "Artificial Intelligence: A Guide for Thinking Humans" - Melanie Mitchell

Online Resources:

- AI Ethics Guidelines: <https://ethics-of-ai.mooc.fi>
- Fairness Indicators:
https://www.tensorflow.org/tfx/guide/fairness_indicators
- NIST AI Risk Management Framework
- Partnership on AI: <https://partnershiponai.org>

Research:

- ACM Conference on Fairness, Accountability, and Transparency (FAccT)
- NeurIPS, ICML workshops on responsible AI

Thank You!

Questions?